

FRACTAL ANTENNA DESIGN FOR WIRELESS POWER TRANSFER APPLICATION

B.E. Final Year Project Abstract
Department of Electronics and Communication Engineering

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Abstract

Wireless Power Transfer (WPT) systems require compact, efficient, and high-performance antennas to harvest electromagnetic energy effectively. Traditional microstrip patch antennas suffer from relatively large physical dimensions and narrow operational bandwidths, which restrict their integration into modern, miniaturized sensor nodes and biomedical implants. This project presents the design, numerical optimization, and physical characterization of a Hexagonal Fractal Microstrip Patch Antenna operating in the 2.4 GHz Industrial, Scientific, and Medical (ISM) band for WPT rectenna configurations.

The proposed design utilizes a regular hexagonal patch substrate printed on a low-cost, fire-retardant FR-4 glass-epoxy dielectric substrate ($\epsilon_r = 4.4$, $h = 1.575$ mm). Miniaturization and impedance matching are achieved by applying fractal geometry through iterative self-similar slotting. Starting with a solid hexagonal patch (Iteration 0), a central hexagonal slot is subtracted (Iteration 1, scale factor $1/3$), followed by nesting six secondary hexagonal slots along the outer vertices (Iteration 2, scale factor $1/9$). This slotting technique forces the surface currents to flow along a longer convoluted perimeter, effectively lengthening the electrical path without increasing the physical outline. Numerical modeling in Autodesk Fusion and 3D electromagnetic simulations in ANSYS HFSS demonstrate that the proposed design achieves an exceptional return loss (S_{11}) of -44.0 dB at 2.4 GHz. The Voltage Standing Wave Ratio (VSWR) is optimized to 1.13, ensuring near-perfect impedance matching to a standard 50-ohm microstrip feedline. The antenna features a symmetrical directional E-plane and a near-omnidirectional H-plane radiation pattern with a peak gain of 2.1 dBi. Fabricated prototypes confirm the simulation findings, demonstrating a 32% overall footprint reduction compared to conventional rectangular patches, making this design highly suitable for compact WPT applications.